TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

AI.206

Computer Architecture

COURSE OVERVIEW

How computers are built: processors, memory hierarchy, I/O, and performance—basis for embedded AI hardware.

1

SESSION

60 MINUTES

ORIENTATION: MACHINES THAT COMPUTE

KEY TOPICS

- Von Neumann architecture
- Instruction cycle
- CPU vs. GPU vs. TPU
- Architecture and AI workloads

TEACHING FOCUS

Understand the machine that runs the software.

KEY TAKEAWAY Architecture determines what software can do.

2

SESSION

60 MINUTES

PROCESSOR DESIGN

KEY TOPICS

- Datapath and control
- Pipelining
- Superscalar and out-of-order
- Instruction sets (RISC/CISC)

TEACHING FOCUS

Execute instructions efficiently.

KEY TAKEAWAY Pipelining lets many instructions overlap.

3

SESSION

60 MINUTES

MEMORY HIERARCHY

KEY TOPICS

- Cache levels and locality
- Virtual memory
- RAM and storage
- Memory bandwidth

TEACHING FOCUS

Feed the processor fast enough.

KEY TAKEAWAY Memory is the bottleneck of modern computing.

4

SESSION

60 MINUTES

I/O & INTERCONNECTS

KEY TOPICS

- Buses and interfaces
- DMA
- Interrupts vs. polling
- GPU interconnects

TEACHING FOCUS

Move data between processor and devices.

KEY TAKEAWAY Fast processors need fast I/O.

5

SESSION

60 MINUTES

PERFORMANCE & PARALLELISM

KEY TOPICS

- Amdahl's law
- Multi-core and many-core
- SIMD and vectorization
- Benchmarking

TEACHING FOCUS

Measure and improve system speed.

KEY TAKEAWAY Parallelism is the path to performance.

6

SESSION

60 MINUTES

ARCHITECTURE FOR AI & REVIEW

KEY TOPICS

- AI accelerators and TPUs
- Edge AI hardware
- Course recap and assessment
- Toward robot design (AI.411)

TEACHING FOCUS

Match hardware to intelligent workloads.

KEY TAKEAWAY AI needs specialized silicon.